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Association Between Periodontitis and Cognitive Impairment or Neurodegenerative Disease: A Systematic Review and Meta-Analysis

Abstract

Objectives: This review aimed to explore whether periodontal disease is linked to cognitive impairment and neurodegenerative conditions in adults and older people.

Materials and Methods: Studies were searched from PubMed/MEDLINE, Scopus, and Web of Science, focusing on observational research published between 2017 and 2025. The review included studies involving adult or older adult populations, with periodontitis or periodontal disease as the main exposure, and cognitive impairment, dementia, Alzheimer disease, Parkinson disease, or other neurodegenerative conditions as outcomes. The reported effect estimates were prepared for pooled analysis and examined using a random-effects model in JASP. Study quality was assessed using the Joanna Briggs Institute critical appraisal tools.

Results: Twelve studies were included, covering 7,335,538 participants from different study designs and data sources. The pooled standardized effect estimate was 0.15, with a 95% confidence interval of 0.05–0.26 and $p=0.007$. The analysis showed substantial heterogeneity, with $Q(11)=39.24$, $p<0.001$, and $I^2=72.0\%$. Sensitivity analysis suggested that the overall result was not strongly influenced by any single study.

Conclusion: Current evidence suggests that periodontal disease may be related to cognitive impairment and neurodegenerative outcomes. However, this relationship should be interpreted carefully because the included studies used different designs, periodontal definitions, and outcome measures. Further longitudinal studies with more consistent periodontal and cognitive assessments are still needed.

Keywords: Alzheimer Disease; Cognitive Dysfunction; Dementia; Neurodegenerative Diseases; Periodontitis

Introduction

Periodontitis is a chronic inflammatory condition that affects the tissues supporting the teeth.

It develops when long-term microbial irritation interacts with the host immune response, gradually leading to periodontal tissue damage. Clinically, this condition may appear as gingival inflammation, periodontal pocket formation, clinical attachment loss, alveolar bone destruction, and tooth loss in more advanced stages. Given the global burden of severe periodontal disease, its possible connection with health problems beyond the oral cavity has become an important area of research [1].

Cognitive impairment and neurodegenerative diseases are common health concerns among adults and older individuals. Dementia is characterized by a gradual decline in memory, thinking ability, behavior, and daily functioning, with Alzheimer disease representing one of its most common forms [2]. Other neurological conditions, such as vascular dementia and Parkinson disease, also contribute to declining function. Because periodontal disease and cognitive decline often occur in aging populations, the possible relationship between oral health and brain health deserves further attention.

Several biological pathways may help explain this possible connection. Periodontal infection may increase systemic inflammation through cytokines and acute-phase mediators, including interleukin-1beta, interleukin-6, tumor necrosis factor-alpha, and C-reactive protein. These inflammatory responses may contribute to endothelial dysfunction, oxidative stress, blood-brain barrier disruption, and neuroinflammation. This concept is often discussed through the

oral-brain axis, which suggests that periodontal pathogens and inflammatory responses may influence pathological changes in the central nervous system [3].

Among the periodontal pathogens, *Porphyromonas gingivalis* has received particular attention because of its virulence factors, including lipopolysaccharides and gingipains [4]. Experimental studies suggest that *P. gingivalis*-induced periodontal inflammation may be accompanied by neuroinflammation, neuronal injury, and dysfunction of the brain barrier [5]. Although these findings do not prove a clinical causal pathway, they provide biological plausibility for examining whether chronic periodontal inflammation is associated with cognitive or neurodegenerative outcomes.

Previous reviews and longitudinal studies have shown that poor periodontal health indicators, such as tooth loss, deep periodontal pockets, and alveolar bone loss, may be associated with cognitive decline, dementia, and Alzheimer disease pathology [6-9]. However, the evidence is not uniform across all neurological outcomes. Findings for Parkinson disease and other neurodegenerative disorders appear less consistent, and interpretation is complicated by differences in periodontal definitions, cognitive outcome measurements, study design, and adjustment for confounding factors [10-12].

A key issue in the current evidence is the diversity of exposure measures, comparison groups, and neurological outcomes. Periodontitis, periodontal treatment status, clinical attachment loss, probing pocket depth, alveolar bone loss, and tooth loss are related indicators, but they do not represent the same clinical exposure. Likewise, mild cognitive impairment, dementia, Alzheimer disease, Parkinson disease, vascular dementia, and amyloid pathology are different clinical outcomes. Therefore, this systematic review and meta-analysis aimed to examine the association between periodontitis and cognitive impairment or

neurodegenerative disease, while considering the clinical and methodological differences among the included studies.

Materials and Methods

Study Design

This study was conducted as a systematic review and meta-analysis following the PRISMA 2020 guideline [15]. It reviewed observational studies that assessed the relationship between periodontal disease and cognitive or neurodegenerative outcomes. The quantitative synthesis and assessment of heterogeneity were performed according to general recommendations for systematic reviews and meta-analyses [16].

PECO Framework

The research question was arranged using the PECO framework. The population included adults and older adults whose periodontal status and cognitive or neurological outcomes were assessed. Tooth loss was not considered as the population, but it was included as a periodontal-related indicator when the original study clearly connected it with periodontal disease or periodontal health. The exposure was periodontitis or periodontal disease. This covered clinical diagnosis, clinical attachment loss, probing pocket depth, bleeding on probing, alveolar bone loss, periodontal treatment status, or other periodontal indicators reported in the included studies. These indicators were treated as related measures of periodontal status, although they were not identical. The comparison group consisted of individuals without periodontitis, individuals with less severe periodontal disease, individuals with better periodontal health, or reference groups defined in the original studies. The outcomes included cognitive impairment, mild cognitive impairment, cognitive decline,

dementia, Alzheimer disease, Parkinson disease, vascular dementia, and other neurodegenerative outcomes. The PECO framework is presented in Table 1.

Search Strategy

The literature search was carried out through PubMed/MEDLINE, Scopus, and Web of Science. The final search was performed in December 2025, and studies published from 2017 to 2025 were considered for inclusion. Search terms were combined using Boolean operators. For periodontal exposure, the keywords included periodontitis, periodontal disease, periodontal health, clinical attachment loss, probing pocket depth, alveolar bone loss, and tooth loss. For cognitive and neurodegenerative outcomes, the keywords included cognitive impairment, cognitive decline, mild cognitive impairment, dementia, Alzheimer disease, Parkinson disease, and neurodegenerative disease. Statistical terms such as odds ratio, confidence interval, risk factor, and association were also used to identify studies suitable for quantitative synthesis. The number of articles identified from each database is shown in Table 2.

Eligibility Criteria

Studies were included when they met the following criteria: published between 2017 and 2025; used a primary quantitative observational design; involved adult or older adult participants; examined periodontitis or periodontal disease as the main exposure; and reported cognitive impairment, cognitive decline, dementia, Alzheimer disease, Parkinson disease, or other neurodegenerative outcomes. Studies also had to provide odds ratios, hazard ratios, confidence intervals, p-values, or sufficient data for effect size calculation. Only full-text articles written in English or Indonesian were considered.

Studies were excluded if they were reviews, systematic reviews, meta-analyses, scoping reviews, narrative reviews, editorials, letters, protocols, case reports, or case series. Articles were also excluded when periodontitis data could not be separated from other oral health exposures, when the study focused only on periodontal microbiology without cognitive or neurological outcomes, when statistical data were insufficient for analysis, or when the same dataset was duplicated across publications.

Study Selection

The selection process was conducted in several stages. After duplicate records were removed, the remaining records were screened by title and abstract. Articles that appeared relevant were then assessed in full text. Studies that did not meet the eligibility criteria were excluded with documented reasons. Any differences in selection decisions were resolved through discussion among the reviewers, and the overall process was reported using the PRISMA flow diagram.

Data Extraction

Data from each eligible article were collected using a standardized extraction form. The information recorded included the author, publication year, country or data source, study design, sample size, periodontal exposure, comparison group, cognitive or neurodegenerative outcome, method of outcome assessment, statistical measure, effect estimate, confidence interval, p-value, follow-up period for cohort studies, and the main finding of the study.

The extraction process was carried out by five reviewers. Effect measures were taken as reported in each original article and included odds ratios, adjusted odds ratios, hazard ratios, adjusted hazard ratios, beta coefficients, confidence intervals, and p-values.

Risk of Bias Assessment

Risk of bias was evaluated with the Joanna Briggs Institute critical appraisal tools, selected according to the design of each included study. The appraisal covered several aspects, including whether the study design was appropriate, how participants were selected, how exposure and outcomes were measured, how confounding factors were addressed, and whether the statistical analysis was suitable. The results of this assessment were used to guide the interpretation of the evidence, not as a basis for automatically excluding studies.

Data Analysis

The meta-analysis was performed in JASP. The main quantitative inputs were the effect estimates and their 95% confidence intervals from the included studies. Reported measures, including odds ratios, adjusted odds ratios, hazard ratios, and adjusted hazard ratios, were extracted based on how they were presented in each original article.

Because the included studies reported different types of effect measures, these estimates were transformed before pooling to allow a more comparable analysis across studies. The final pooled result was then reported as a standardized effect size.

The pooled estimate was generated using a random-effects model because differences were expected among the included studies. These differences involved study design, definition of periodontal exposure, comparison group, cognitive or neurodegenerative outcome, and type of statistical measure. For this reason, the pooled result was interpreted as a general association and not as evidence of a causal relationship.

Heterogeneity was examined using the Q statistic and I² value. The Q statistic was used to determine whether variation between studies was statistically significant, while I² described how much of the variation was due to real differences between studies rather than chance. The I² value was 72.0%, showing substantial heterogeneity.

Sensitivity analysis was conducted to check whether the pooled estimate was strongly affected by any single study. The result showed that no individual study had a major influence on the overall finding. A funnel plot was also used to visually explore possible publication bias. However, this assessment was interpreted carefully because only 12 studies were included. A p-value below 0.05 was considered statistically significant.

Ethical Considerations

This review used only data from studies that had already been published. No participants were recruited directly, and no new biological samples or animal experiments were involved. For this reason, ethical approval and informed consent were not required.

Results

Study Selection

The selection process was carried out according to the PRISMA 2020 guideline [15]. The database search identified 378 records in total, consisting of 124 records from PubMed/MEDLINE, 156 from Scopus, and 98 from Web of Science. After 5 duplicate records were removed, 373 records were screened by title and abstract.

During the initial screening stage, 291 records were excluded. The remaining 82 articles were then reviewed in full text, and 70 articles were excluded for specific reasons. In the final stage, 12 studies fulfilled the eligibility criteria and were included in both the qualitative review and quantitative meta-analysis. Overall, these studies represented 7,335,538 participants from various study designs and data sources. The PRISMA flow diagram is presented in Figure 1.

Characteristics of the Included Studies

The 12 studies included in this review were published from 2017 to 2025. The largest number of articles was published in 2017 and 2025, with three studies in each of these years. In terms of country or data source, most studies came from Taiwan, South Korea, and the United States, while the remaining studies were conducted in Mexico, Japan, and Spain.

Regarding study design, six studies used cohort or population-based cohort designs, and the other six used cross-sectional or observational analytical approaches. The outcomes most often assessed were cognitive impairment, mild cognitive impairment, and amyloid pathology. Other outcomes included dementia, Alzheimer disease, vascular dementia, and Parkinson disease. A summary of the study characteristics is provided in Table 3.

Summary of Findings

The included studies generally showed a relationship between periodontal disease and cognitive or neurodegenerative outcomes. In the cohort studies, chronic or untreated periodontitis was linked with dementia or Alzheimer disease [17-20,23]. Another large nationwide cohort study found a relationship between periodontitis and Parkinson disease [22].

The cross-sectional and observational studies also showed links between periodontal disease indicators and cognitive-related outcomes. These indicators included clinical attachment loss, severe periodontitis, tooth loss, mild cognitive impairment, cognitive test performance, and beta-amyloid accumulation [21,24-28]. However, the size and precision of the reported effects differed between studies. This variation may be explained by differences in study design, sample size, periodontal definitions, cognitive outcome measurements, comparison groups, and adjustment for confounding factors.

Funnel Plot

3 The funnel plot was used to assess the distribution of studies based on residual values and standard errors. Some studies were located near the central line, whereas others were positioned farther from the center, indicating visual imbalance. This pattern may be related to variation in sample size, study design, outcome category, and the statistical measures used across the included studies. Since the review included only 12 studies, the funnel plot findings were interpreted with caution. The funnel plot is shown in Figure 2.

Forest Plot and Pooled Effect

20 The forest plot showed that most studies had a positive direction of effect. The pooled effect estimate was 0.15, representing a standardized effect size, with a 95% confidence interval of 0.05-0.26 and $p=0.007$. This result indicates a significant relationship between periodontitis and cognitive impairment or neurodegenerative outcomes. Significant heterogeneity was observed, with $Q(11)=39.24$, $p<0.001$, and $I^2=72.0\%$, showing that the included studies varied substantially. This variation needs to be considered when interpreting the pooled result, as the studies differed in participant characteristics, exposure definitions, outcome types, study designs, and statistical approaches. The forest plot is shown in Figure 3.

Discussion

79 This review found an overall positive relationship between periodontitis and cognitive impairment or neurodegenerative outcomes. The pooled standardized effect estimate was 0.15, with a 95% confidence interval of 0.05-0.26 and $p=0.007$. Although this result indicates statistical significance, it should not be interpreted as evidence of causation because the included studies varied in exposure definitions, outcome categories, research designs, comparison groups, and statistical measures.

3 The present findings are broadly consistent with previous studies reporting links between
25 40 poor periodontal health, cognitive decline, and dementia [6-9]. In the cohort studies included
in this review, dementia or Alzheimer disease was more often reported among individuals
with chronic, untreated, or severe periodontitis [17-20,23]. Cross-sectional studies also
reported relationships between periodontal parameters and cognitive outcomes [21,24-28].
Nevertheless, the magnitude of these associations differed across studies, and the temporal
direction of the relationship remains uncertain in non-longitudinal designs.

Possible explanatory mechanisms include inflammatory, microbial, vascular, and immune
pathways. Periodontal pathogens and the inflammatory response to periodontal infection may
contribute to systemic inflammation, vascular dysfunction, oxidative injury, and
neuroinflammatory processes. 78 The oral-brain axis and the potential involvement of P.
gingivalis have been proposed as biologically plausible links [3-5]. However, mechanistic
plausibility does not establish that periodontal disease causes cognitive decline, and
epidemiological results must be considered alongside confounding, reverse causality, and
measurement variability.

The strength of association may also differ by neurological outcome. The evidence in this
review appeared more consistent for dementia and Alzheimer disease, whereas Parkinson
disease was represented by only one included study. Mild cognitive impairment and beta-
amyloid accumulation may reflect earlier or more specific stages of cognitive deterioration,
but they are not equivalent to dementia or Parkinson disease. Combining these outcomes in
a single pooled estimate improves overall synthesis but reduces clinical specificity and may
contribute to heterogeneity.

Substantial heterogeneity was observed across the meta-analysis. The included studies used different periodontal exposure definitions, including clinical periodontitis, chronic periodontitis, severe periodontitis, clinical attachment loss, periodontal treatment status, and tooth loss related to periodontal status. Outcome assessment also varied and included clinical diagnoses, registry data, medical records, cognitive tests, and imaging-based measures. These differences likely influenced effect estimates and should be considered when interpreting the pooled result.

Residual confounding and reverse causation remain important limitations. Factors such as age, sex, education, smoking, diabetes, socioeconomic position, depression, cardiovascular disease, oral hygiene behavior, access to dental care, and general health status may be associated with both periodontal disease and cognitive outcomes. Although several primary studies adjusted for selected confounders, unmeasured or incompletely measured factors may still have affected the observed associations. Cognitive impairment may also lead to poorer oral hygiene and less frequent dental care, which can worsen periodontal status after cognitive decline has already started.

Publication bias should be interpreted cautiously. A funnel plot was included to provide a visual assessment, but the small number of included studies limits the reliability of any judgment about asymmetry. Differences in sample size, study design, outcome type, and effect measure may also influence the appearance of the funnel plot. Therefore, the possibility that publication bias affected the overall evidence cannot be excluded.

This review has several limitations. First, most included studies were observational, which limits causal interpretation. Second, definitions of periodontal exposure were not consistent across studies. Third, cognitive and neurodegenerative outcomes were assessed using

45 different diagnostic approaches and instruments. Fourth, the quantitative synthesis combined several statistical measures, including odds ratios, adjusted odds ratios, hazard ratios, adjusted hazard ratios, and beta coefficients. Fifth, heterogeneity was significant, and subgroup analyses by outcome type, study design, exposure definition, or statistical measure were not performed. Sixth, certainty of evidence was not formally evaluated using the GRADE framework. Finally, limiting the search to studies published from 2017 to 2025 may have omitted relevant earlier evidence.

Future research should prioritize longitudinal designs with longer follow-up, standardized periodontal definitions, repeated cognitive assessments, and more comprehensive adjustment for confounding variables. Future reviews may also benefit from separate syntheses by neurological outcome, study design, exposure definition, and effect measure rather than combining clinically distinct outcomes into one broad estimate. Additional studies incorporating inflammatory biomarkers, periodontal pathogen profiles, and standardized neurocognitive assessment may help clarify the biological pathways connecting periodontal disease with cognitive and neurodegenerative outcomes.

Conclusion

26 The current evidence suggests that periodontitis and periodontal disease are linked with cognitive impairment and neurodegenerative outcomes. However, this finding should be understood as an epidemiological association, not as proof that periodontal disease causes or prevents these conditions. Because the included studies differed in clinical characteristics and methods, the pooled estimate needs to be interpreted carefully. Further longitudinal studies with standardized periodontal and cognitive assessments, stronger control of

2 confounding factors, and separate analyses for each neurological outcome are needed to better understand the direction and strength of this relationship.

Declaration of Artificial Intelligence (AI) Use

11 The authors used ChatGPT (OpenAI) to support language editing, improve sentence flow, and enhance the clarity of the manuscript. 4 The tool was not used to create or modify study data, determine screening or eligibility decisions, assess risk of bias, perform statistical analysis, interpret the findings, or formulate the conclusions. 1 All AI-assisted text was checked, revised, and confirmed by the authors, who remain fully responsible for the final manuscript content.

References

1. World Health Organization. Oral health [Internet]. Geneva: World Health Organization; 2025 [cited 2026 Jun 5]. Available from: <https://www.who.int/news-room/fact-sheets/detail/oral-health>
2. World Health Organization. Dementia [Internet]. Geneva: World Health Organization; 2025 [cited 2026 Jun 5]. Available from: <https://www.who.int/news-room/fact-sheets/detail/dementia>
3. Li R, Wang J, Xiong W, Luo Y, Feng H, Zhou H, et al. The oral-brain axis: can periodontal pathogens trigger the onset and progression of Alzheimer's disease? *Front Microbiol.* 2024;15:1358179. doi:10.3389/fmicb.2024.1358179.
4. Huang Z, Hao M, Shi N, Wang X, Yuan L, Yuan H, et al. *Porphyromonas gingivalis*: a potential trigger of neurodegenerative disease. *Front Immunol.* 2025;16:1482033. doi:10.3389/fimmu.2025.1482033.

5. Jiang Y, Xu L, Zhao X, Shen H, Qiu C, He Z, et al. Porphyromonas gingivalis-induced periodontitis promotes neuroinflammation and neuronal loss associated with dysfunction of the brain barrier. *Front Cell Infect Microbiol.* 2025;15:1559182. doi:10.3389/fcimb.2025.1559182.
6. Asher S, Stephen R, Mäntylä P, Suominen AL, Solomon A. Periodontal health, cognitive decline, and dementia: a systematic review and meta-analysis of longitudinal studies. *J Am Geriatr Soc.* 2022;70(9):2695-2709. doi:10.1111/jgs.17978.
7. Said-Sadier N, Sayegh B, Farah R, Abbas LA, Dweik R, Tang N, et al. Association between periodontal disease and cognitive impairment in adults. *Int J Environ Res Public Health.* 2023;20:4707. doi:10.3390/ijerph20064707.
8. Dibello V, Custodero C, Cavalcanti R, Lafornera D, Dibello A, Lozupone M, et al. Impact of periodontal disease on cognitive disorders, dementia, and depression: a systematic review and meta-analysis. *GeroScience.* 2024;46(5):5133-5169. doi:10.1007/s11357-024-01243-8.
9. Fu YD, Li CL, Hu CL, Pei MD, Cai WY, Li YQ, et al. Meta analysis of the correlation between periodontal health and cognitive impairment in the older population. *J Prev Alzheimers Dis.* 2024;11. doi:10.14283/jpad.2024.87.
10. Plachokova AS, Gjaltema J, Hagens ERC, Hashemi Z, Knüppe TBA, Kootstra TJM, et al. Periodontitis: a plausible modifiable risk factor for neurodegenerative diseases? A comprehensive review. *Int J Mol Sci.* 2024;25(8):4504. doi:10.3390/ijms25084504.
11. Villar A, Paladini S, Cossatis J. Periodontal disease and Alzheimer's: insights from a systematic literature network analysis. *J Prev Alzheimers Dis.* 2024;11(4):1148-1165. doi:10.14283/jpad.2024.79.

12. Barbarisi A, Visconti V, Lauritano D, Cremonini F, Caccianiga G, Ceraulo S. Correlation between periodontitis and onset of Alzheimer's disease: a literature review. *Dent J*. 2024;12(10):331. doi:10.3390/dj12100331.
13. Erlianti CP, Trihandini I. Factors associated with cognitive impairment among Indonesian older adults. *Aging Med Healthc*. 2024;15(3):105-113. doi:10.33879/AMH.153.2022.05048.
14. van der Ploeg ES, Handajani YS, Schröder-Butterfill E. Dementia care in Indonesia: care networks, awareness & perception. *J Cross Cult Gerontol*. 2025;40(4):441-461. doi:10.1007/s10823-025-09544-x.
15. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. doi:10.1136/bmj.n71.
16. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editors. *Cochrane handbook for systematic reviews of interventions*. Version 6.5. Cochrane; 2024. Available from: <https://training.cochrane.org/handbook>
17. Chen CK, Wu YT, Chang YC. Association between chronic periodontitis and the risk of Alzheimer's disease: a retrospective, population-based, matched-cohort study. *Alzheimers Res Ther*. 2017;9:56. doi:10.1186/s13195-017-0282-6.
18. Lee YT, Lee HC, Hu CJ, Huang LK, Chao SP, Lin CP, Su ECY, et al. Periodontitis as a modifiable risk factor for dementia: a nationwide population-based cohort study. *J Am Geriatr Soc*. 2017;65(2):301-305. doi:10.1111/jgs.14449.

19. Lee YL, Hu HY, Huang LY, Chou P, Chu DK. Periodontal disease associated with higher risk of dementia: population-based cohort study in Taiwan. *J Am Geriatr Soc.* 2017;65(9):1975-1980. doi:10.1111/jgs.14944.
20. Choi S, Kim K, Chang J, Kim SM, Kim SJ, Cho HJ, et al. Association of chronic periodontitis on Alzheimer's disease or vascular dementia. *J Am Geriatr Soc.* 2019;67(6):1234-1239. doi:10.1111/jgs.15828.
21. Sung CE, Huang RY, Cheng WC, Kao TW, Chen WL. Association between periodontitis and cognitive impairment: analysis of National Health and Nutrition Examination Survey III. *J Clin Periodontol.* 2019;46(8):790-798. doi:10.1111/jcpe.13155.
22. Jeong E, Park JB, Park YG. Evaluation of the association between periodontitis and risk of Parkinson's disease: a nationwide retrospective cohort study. *Sci Rep.* 2021;11:16594. doi:10.1038/s41598-021-96147-4.
23. Kim SR, Son M, Kim YR, Kang HK. Risk of dementia according to the severity of chronic periodontitis in Korea: a nationwide retrospective cohort study. *Epidemiol Health.* 2022;44:e2022077. doi:10.4178/epih.e2022077.
24. Luo H, Wu B, González HM, Stickel A, Kaste LM, Tarraf W, et al. Tooth loss, periodontal disease, and mild cognitive impairment among Hispanic/Latino immigrants: the moderating effects of age at immigration. *J Gerontol A Biol Sci Med Sci.* 2023;78(6):949-957. doi:10.1093/gerona/glac178.
25. Igase M, Igase K, Hino S, Uchida D, Okada Y, Ochi M, et al. Association of periodontitis with mild cognitive impairment in older adults. *J Aging Res Lifestyle.* 2024;13:108-112. doi:10.14283/jarlife.2024.16.

26. Gil-Montoya JA, Gerez-Muñoz MJ, Triviño-Ibáñez E, Carrera-Muñoz I, Bravo M, Rashki M, et al. Periodontal disease and brain amyloid pathology in mild cognitive impairment. *Neurologia*. 2025;40(7):641-650. doi:10.1016/j.nrleng.2025.07.008.
27. Sun Q, Li M. Association between periodontitis and cognitive impairment in older adults: a cross-sectional study of the National Health and Nutrition Examination Survey. *Clin Epidemiol Glob Health*. 2025;33:102020. doi:10.1016/j.cegh.2025.102020.
28. Aranda Romo S, Garrocho-Rangel A, Marín González A, Cepeda-Bravo A, Jiménez-Capdeville ME, Ortega Rodríguez C, et al. Cognitive status and periodontitis: is there a correlation? A cross-sectional study. *Dent Med Probl*. 2025;62(4):591-600. doi:10.17219/dmp/177412.

Table 1. Research PECO

Component	Description
P (Population)	Adult or older adult individuals whose periodontal status and cognitive or neurological status were assessed.
E (Exposure)	Periodontitis or periodontal disease, including clinical diagnosis, clinical attachment loss, probing pocket depth, bleeding on probing, alveolar bone loss, periodontal treatment status, or other periodontal indicators reported in the included studies. These exposure definitions were considered heterogeneous.
C (Comparison)	Individuals without periodontitis, individuals with less severe periodontal disease, individuals with better periodontal health, or reference groups defined in the original studies. These comparison groups were not identical across studies.
O (Outcome)	Cognitive impairment, mild cognitive impairment, cognitive decline, dementia, Alzheimer's disease, Parkinson's disease, and other neurodegenerative diseases assessed through clinical diagnosis, registry data, medical records, or standardized cognitive instruments.

Table 2. Number of Articles Based on Search Databases

Database	Keyword	Number of Articles
PubMed/MEDLINE	(“periodontitis” OR “periodontal disease” OR “periodontal health” OR “clinical attachment loss” OR “probing pocket depth” OR “alveolar bone loss” OR “tooth loss”) AND (“cognitive impairment” OR “cognitive decline” OR “mild cognitive impairment” OR dementia OR “Alzheimer disease” OR “Parkinson disease” OR “neurodegenerative disease”) AND (“odds ratio” OR “hazard ratio” OR “confidence interval” OR “p-value” OR association)	124
Scopus	TITLE-ABS-KEY (“periodontitis” OR “periodontal disease” OR “periodontal health” OR “tooth loss” OR “alveolar bone loss”) AND TITLE-ABS-KEY (“cognitive impairment” OR “cognitive decline” OR dementia OR “Alzheimer disease” OR “Parkinson disease” OR “neurodegenerative disease”) AND TITLE-ABS-KEY (“odds ratio” OR “confidence interval” OR association OR “risk factor”) AND PUBYEAR > 2016	156
Web of Science	TS=(“periodontitis” OR “periodontal disease” OR “periodontal health” OR “clinical attachment loss” OR “probing pocket depth” OR “tooth loss”) AND TS=(“cognitive impairment” OR “cognitive decline” OR dementia OR “Alzheimer disease” OR “Parkinson disease” OR “neurodegenerative disease”) AND TS=(“odds ratio” OR “confidence interval” OR association OR “risk factor”)	98
Total articles identified	Total articles identified	378

Table 3. Summary of SLR Articles on the Association of Periodontitis With Cognitive Impairment and Neurodegenerative Disease

No.	Author(s) (Year)	Objective	Method	Results	Conclusion
1	Chen et al. (2017)	To analyze the risk of Alzheimer's disease in patients with chronic periodontitis.	A retrospective population-based matched cohort study in Taiwan. The sample consisted of 9,291 patients with chronic periodontitis and 18,672 controls without chronic periodontitis. The analysis used Cox proportional hazards regression.	Patients with 10 years of exposure to chronic periodontitis had a higher risk of Alzheimer's disease than those without chronic periodontitis. The adjusted HR was 1.707; 95% CI = 1.152–2.528; $p = 0.0077$.	Long-term exposure to chronic periodontitis is associated with an increased risk of Alzheimer's disease. These findings suggest that periodontal status may be relevant to neurodegenerative outcomes, although causality cannot be inferred.
2	Y.-T. Lee et al. (2017)	To assess whether periodontitis can serve as a modifiable risk factor for incident dementia.	A prospective cohort study using the Taiwan National Health Insurance Research Database. The sample consisted of 3,028 older adults with periodontitis and 3,028 age- and sex-matched controls.	The risk of dementia was higher in the periodontitis group than in the control group. The HR was 1.16; 95% CI = 1.01–1.32; $p = 0.03$.	Periodontitis is associated with the risk of new-onset dementia. Periodontal infection may be considered a potentially modifiable oral health factor, although preventive effects require further evidence.
3	Y.-L. Lee et al. (2017)	To assess the magnitude and temporal aspects of the association between periodontal disease and dementia risk.	A retrospective cohort study based on the Taiwan National Health Insurance Research Database. The sample included 182,747 individuals with newly diagnosed periodontal disease. The analysis compared dental prophylaxis, intensive periodontal	The incidence of dementia was higher in the untreated periodontal disease group at 0.76% per year and in the tooth extraction group at 0.57% per year. The untreated group had an HR of 1.14; 95% CI = 1.04–1.24. The tooth extraction group had an HR of 1.10; 95% CI = 1.04–1.16. Differences in incidence between groups were	More severe or untreated periodontal disease is associated with a higher risk of dementia. Periodontal treatment status may be related to differences in dementia incidence, although causality cannot be inferred.

No.	Author(s) (Year)	Objective	Method	Results	Conclusion
			treatment, tooth extraction, and no treatment groups.	significant, with $p < 0.001$.	
4	Choi et al. (2019)	To analyze the association between chronic periodontitis and overall dementia, Alzheimer's disease, and vascular dementia.	A retrospective cohort study based on the Korean National Health Insurance Service-Health Screening Cohort. The sample consisted of 262,349 participants. The analysis used Cox proportional hazards regression.	Chronic periodontitis was associated with an increased risk of overall dementia, with an aHR of 1.06; 95% CI = 1.01–1.11; $p = 0.015$. The risk of Alzheimer's disease also increased, with an aHR of 1.05; 95% CI = 1.00–1.11; $p = 0.042$. The risk of vascular dementia increased but was not statistically significant, with an aHR of 1.10; 95% CI = 0.98–1.22; $p = 0.088$.	Chronic periodontitis is associated with a higher risk of dementia and Alzheimer's disease. The association with vascular dementia appears to increase but is not statistically significant.
5	Sung et al. (2019)	To analyze the association between periodontitis and impaired cognitive function in the adult population of the United States.	A cross-sectional study using NHANES III. The sample consisted of 4,663 participants aged 20–59 years. Periodontitis status was categorized as normal, mild, moderate, and severe. Cognitive function was assessed using SRTT, SDST, and SDLT.	After adjustment for demographic factors, education, smoking, cardiovascular disease, and laboratory data, periodontitis was significantly associated with increased SDST and SDLT scores. The p for trend was 0.014 for SDST and 0.038 for SDLT. In the fully adjusted model, moderate-to-severe periodontitis was associated with the SDLT total score, with $\beta = 0.146$; 95% CI = 0.010–0.282; $p = 0.036$.	Periodontal status is associated with reduced cognitive function in a national adult sample. Individuals with cognitive impairment require attention in periodontal care.
6	Jeong et al. (2021)	To evaluate the association between periodontitis and the risk of Parkinson's disease.	A nationwide retrospective cohort study using data from the South Korean National Health Insurance Service. The final sample consisted of 6,825,684 participants aged	The risk of Parkinson's disease increased among individuals requiring further dental visits, with an HR of 1.142; 95% CI = 1.094–1.193. The group with periodontitis and a need for further dental visits had the highest	Periodontitis is associated with an increased risk of Parkinson's disease. Regular dental and periodontal care may be relevant to broader health management, although its effect

No.	Author(s) (Year)	Objective	Method	Results	Conclusion
			≥40 years. The analysis used the Cox proportional hazards model.	risk, with an HR of 1.167; 95% CI = 1.118–1.219.	on neurodegenerative disease risk remains unconfirmed.
7	Kim et al. (2022)	To analyze the risk of dementia among older adults with chronic periodontitis according to the severity of periodontitis.	A nationwide retrospective cohort study based on the South Korean National Health Insurance Service-Senior cohort database. After propensity score matching, there were 8,624 patients with mild chronic periodontitis and 8,624 patients with severe chronic periodontitis.	The severe chronic periodontitis group had a higher risk of dementia than the mild chronic periodontitis group. The HR was 1.15; 95% CI = 1.04–1.27; p = 0.009. Kaplan-Meier analysis showed a lower disease-free probability in the severe chronic periodontitis group, with p for log-rank = 0.001.	The severity of chronic periodontitis is associated with an increased risk of dementia. Management of periodontitis from the mild stage may be clinically relevant, although its role in altering dementia risk requires further study.
8	Luo et al. (2023)	To assess the association of tooth loss and periodontal disease with mild cognitive impairment among Hispanic/Latino immigrants and to examine the role of age at immigration.	An observational study using data from the Hispanic Community Health Study/Study of Latinos. The analyzed immigrant sample consisted of 5,709 participants. The analysis used adjusted regression models.	Severe tooth loss was associated with MCI, with an AOR of 1.36; 95% CI = 1.01–1.85. Overall, periodontal disease was associated with MCI, with p = 0.04. Comparisons of mild, moderate, and severe periodontal disease with no periodontal disease were not significant, with all p values > 0.05. The interaction between periodontal disease and age at immigration was also not significant, with p = 0.10.	Tooth loss appears to be a stronger oral health indicator of MCI than periodontal disease severity. Periodontal disease still shows a general association with MCI.
9	Igase et al. (2024)	To examine the association between severe periodontitis and mild cognitive impairment among community	A cross-sectional study involving 321 community participants in Japan. MCI was assessed using the MCI screening test. Periodontitis was diagnosed	MCI was found in 41 subjects. Severe periodontitis was found in 123 subjects. Severe periodontitis was significantly associated with MCI after including age, education, number of	Severe periodontitis is strongly associated with mild cognitive impairment. Patients with severe periodontitis

No.	Author(s) (Year)	Objective	Method	Results	Conclusion
		residents who underwent dental health examinations.	based on probing pocket depth. The analysis used multivariate logistic regression.	functional teeth, and the presence of severe periodontitis. The OR was 4.024; 95% CI = 1.768–9.161; $p < 0.001$.	should be considered for cognitive function screening.
10	Gil-Montoya et al. (2025)	To assess the association between periodontal disease and brain beta-amyloid accumulation in patients with mild cognitive impairment.	An observational study in Spain. All participants underwent periodontal examination and amyloid-PET scanning. The sample consisted of MCI/A β + $n = 45$, MCI/A β - $n = 59$, and non-MCI/A β - $n = 60$.	Patients with moderate-to-severe periodontitis had a higher risk of abnormal A β accumulation. The comparison between MCI/A β + and MCI/A β - showed OR = 3.30; 95% CI = 1.30–8.26. The comparison between MCI/A β + and non-MCI/A β - showed OR = 4.94; 95% CI = 1.65–14.84.	Periodontal disease is associated with brain beta-amyloid accumulation. These findings support the suspected link between periodontal disease and Alzheimer's pathology at the MCI stage.
11	Sun & Li (2025)	To analyze the association between periodontitis and cognitive impairment among older adults in the United States.	A cross-sectional study using NHANES 2011–2014. The sample consisted of 2,514 individuals aged ≥ 60 years. Periodontitis was categorized as no/mild and moderate/severe. Cognitive function was assessed using CERAD-WL, AFT, and DSST.	A total of 52.6% of the sample had moderate/severe periodontitis. Moderate/severe periodontitis was associated with cognitive impairment on the DSST. The fully adjusted model showed OR = 1.290; 95% CI = 1.021–1.630; $p = 0.033$. Differences in CERAD-WL and DSST scores across periodontal status were significant, with $p = 0.046$ and $p = 0.010$.	Periodontitis is significantly associated with cognitive impairment among older adults. Periodontal care may be considered within older adult health management, although its effect on cognitive decline requires further study.
12	Aranda Romo et al. (2025)	To determine the correlation between gingival index, periodontitis staging, and mild cognitive impairment assessed using MoCA in adults aged ≥ 55 years.	A cross-sectional study involving 120 Mexican patients with or without periodontal disease. Periodontal examinations were conducted clinically and radiographically.	A total of 54% of subjects had MCI. Loss of clinical attachment level was associated with MCI, with OR = 1.68; 95% CI = 1.06–2.68; $p = 0.027$. Gingival index was also correlated with MCI, with $p = 0.025$. Other associated variables were sex,	Gingival and periodontal parameters, particularly clinical attachment loss, are associated with mild cognitive impairment in older adults. Periodontal

No.	Author(s) (Year)	Objective	Method	Results	Conclusion
			MCI was assessed using the Montreal Cognitive Assessment.	with OR = 2.97; p = 0.015, and education, with OR = 0.37; p = 0.021.	examination may provide additional information for the early detection of cognitive impairment.

Figure Legends

Figure 1. PRISMA Flow Diagram of Study Selection.

The diagram shows the identification, screening, eligibility assessment, and inclusion process of articles in this systematic review and meta-analysis.

Figure 2. Funnel Plot of the Relationship Between Periodontitis, Cognitive

Impairment, and Neurodegenerative Disease.

The funnel plot presents the distribution of included studies based on residual values and standard errors to visually assess possible publication bias and variation across studies.

Figure 3. Forest Plot of the Association Between Periodontitis, Cognitive Impairment, and Neurodegenerative Disease.

The forest plot presents the effect estimate, confidence interval, pooled effect, and heterogeneity of the included studies.